Satellite Image Segmentation with Pretrained Models

Master of Science in AI & Robotics

Neural Networks course

Simone La Bella 1995847

Mattia Di Marco 2019367

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SAPIENZA
UNIVERSITÀ DI ROMA

Motivation and Aim

Problem

- Building footprint segmentation from aerial imagery
- Dense annotations are expensive to obtain
- Accurate boundaries are important

Objectives

- Evaluate **SatMAE++ pretraining**
- Compare against **Swin-Unet**
- Explore **wavelet-based strategies**
- Investigate **SAM pseudo-labeling** with limited annotations

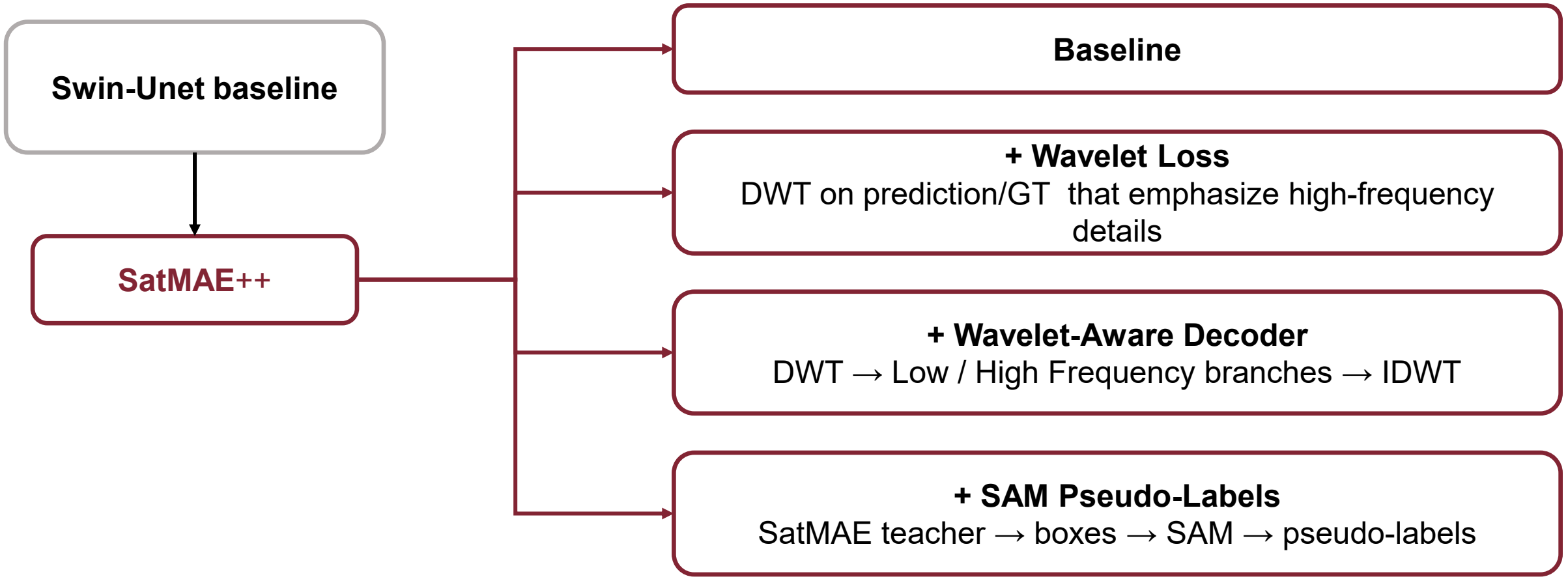


Segmentation Model

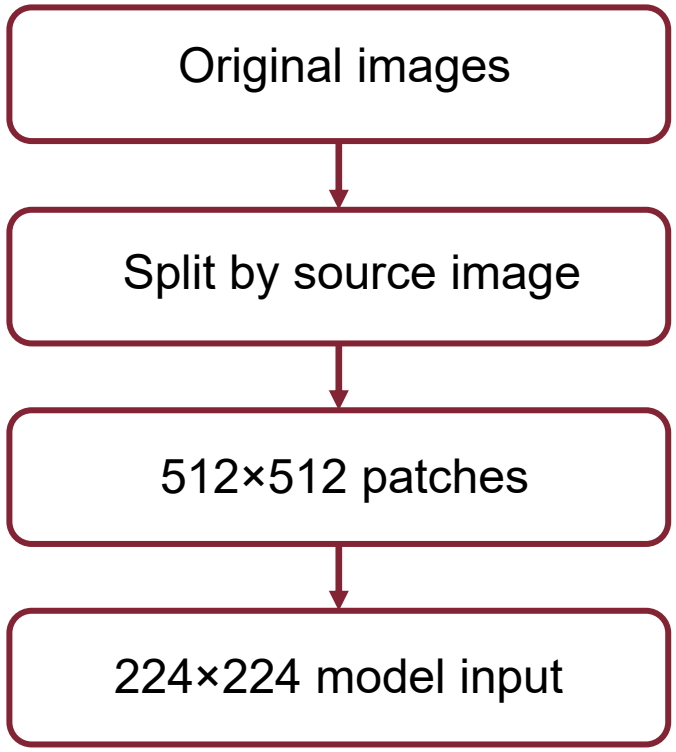


Dataset: INRIA Aerial Image Labeling

Method Overview



Implementation & Experiments



- **14,400** training patches
- **3,600** validation patches

Settings of the training

Input	224x224
Epochs	20
Early Stopping	5
Effective Batch	4
Optimizer	AdamW
Main Loss	BCE + Dice

Evaluation Metrics: IoU, Dice, Precision, Recall, Accuracy



Results

	IoU	Dice	Precision	Recall	Accuracy
Swin-Unet	0.685	0.813	0.800	0.827	0.891
SatMAE++ baseline	0.732	0.845	0.809	0.885	0.907
SatMAE++ Wavelet Loss	0.730	0.844	0.796	0.898	0.905
SatMAE++ Wavelet-Aware Decoder	0.729	0.843	0.851	0.835	0.911
SatMAE++ SAM Pseduo-Labels	0.685	0.813	0.722	0.930	0.877

Results



Original image

Swin-Unet

SatMAE++

SatMAE++ SAM Pseduo-Labels

SatMAE++ Wavelet-Aware Decoder

SatMAE++ Wavelet Loss

Conclusions and limitations

SatMAE++

Best overall model:
- IoU **0.732**
- Remote-sensing pretraining transfers effectively to segmentation

Wavelet

Different precision/recall trade-offs
- Wavelet Loss → Higher Recall
- Wavelet Decoder → Higher Precision
- No improvement in best IoU

SAM

Promising for label-scarce training
- Only **10% real annotations**
- Highest Recall: **0.930**
- More false positives